

Nuclear fission and fusion

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Define nuclear fission.	The splitting of a large, unstable nucleus into two smaller nuclei.	☐
2	Give two examples of elements that can undergo nuclear fission.	Uranium and plutonium	☐
3	What is spontaneous fission?	Fission that happens without the nucleus absorbing a neutron.	☐
4 ★	State what usually has to happen before a nucleus undergoes fission.	The unstable nucleus must absorb a neutron.	☐
5 ★	State what a nucleus splits into during fission.	Two smaller nuclei, two or three neutrons and gamma rays	☐
6 ★	What is a chain reaction?	Neutrons released by one fission cause further fissions, which release more neutrons, and so on.	☐
7 ★	Define nuclear fusion.	The joining of two light nuclei to form a heavier nucleus.	☐
8	In nuclear fusion, which has more mass: the two light nuclei or the nucleus they form?	The two light nuclei	☐
9 ★	Some mass is lost during nuclear fusion. State what it is converted into.	Energy (of radiation)	☐
10	Name the energy store that all fission products have.	The kinetic energy store	☐
11 ★	State how the chain reaction in a reactor differs from that in a nuclear weapon.	Reactor: controlled. Weapon: uncontrolled, causing an explosion.	☐
12	Name the process that releases energy in stars.	Nuclear fusion	☐